Akshat Sharma

EMBEDDED FIRMWARE | FPGA | EMBEDDED LINUX

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EDUCATION

Maharaja Agrasen Institute of Technology — **2023–2027**
B.Tech. in Electronics and Communication Engineering; **CGPA: 8.46/10** — Delhi, India

EXPERIENCE

Embedded Systems Engineering Intern — **Jun 2026–Present**
Rebhu Computing — *Greater Noida, India*

- Developing a plug-and-play **visual–inertial odometry (VIO) navigation device** for GPS-denied UAV operation; the device presents its position estimates to ArduPilot by emulating a **MAVLink-compatible GPS sensor**.
- Implemented **matrix-multiplication accelerator datapaths** on an **Artix-7 100T**, taking designs through RTL development, synthesis, and on-hardware validation for AI inference workloads.
- Engineered the **Bluetooth LE 5.4 PAwR** communication path for electronic shelf-label networks, enabling synchronized, bidirectional exchanges between an access point and many power-constrained tags.
- Developed **SPI display drivers** and debugged the hardware/software boundary during board bring-up, peripheral integration, signal validation, and low-power display-update work.

Open Source Hardware Intern — **Dec 2025**
FOSSEE, IIT Bombay — *Remote*

- Built an **emulation-backed grading pipeline** that compiled and executed Arduino, MicroPython, and ESP-IDF submissions under QEMU, removing the physical-board dependency from Yaksh evaluations.
- Designed the test harness around **bounded execution, stdout capture/filtering, result comparison, and actionable diagnostics**, making firmware assessment deterministic and repeatable at scale.

Robotics Intern — **Jun 2025–Aug 2025**
Rebhu Computing — *Greater Noida, India*

- Integrated **CAN wheel encoders, an IMU, Jetson Orin, and RTSP cameras** into an autonomous UGV; validated the interface and localization stack through ArduPilot–Gazebo SITL.

SELECTED ENGINEERING PROJECTS

WS2812B DMA Driver — *C, STM32F102* — **GitHub**

- Developed a **non-blocking driver for WS2812B LED chains** that generates the timing-critical 800 kHz single-wire protocol while leaving the CPU available for other application work.
- Encoded each LED's 24-bit GRB value into timer-compare data and streamed it to the PWM peripheral using **DMA**; generated the reset interval in the same buffer, guarded overlapping transfers with a completion callback, and verified timing on an oscilloscope.

Bare-Metal STM32 Pong — *C, ARM Cortex-M3* — **GitHub**

- Built a **playable Pong game entirely as bare-metal firmware** on an STM32F103, using an SH1106 OLED for graphics and a rotary encoder for paddle control, without a HAL or RTOS.
- Wrote the startup code, linker script, register-level GPIO/SPI drivers, framebuffer primitives, and 1 ms SysTick input loop; implemented START–PLAY–WIN states, scoring, collision detection, and impact-dependent ball deflection.

CHIP-8 Computer on FPGA — *Verilog, Cyclone IV* — **GitHub**

- Built a **standalone CHIP-8 gaming computer in Verilog** that loads and runs CHIP-8 ROMs directly on the DE0-Nano FPGA without a software processor or operating system.
- Designed the CPU, 4 KB memory, and 64×32 framebuffer; integrated VGA output, PS/2 keyboard input, and timer-driven audio, then simulated the system with Verilator and validated it on Cyclone IV hardware.

ADDITIONAL SYSTEMS WORK

Built a **bare-metal RISC-V boot path** from reset assembly and a custom linker image to MMIO UART under QEMU; designed an **ATmega32U4 USB keypad** with production-ready KiCad/Gerbers; configured a minimal **Buildroot Linux image and SH1106 SPI framebuffer** for Raspberry Pi Zero 2 W.

TECHNICAL SKILLS

Languages: C/C++, Python, Verilog, MicroPython, RISC-V Assembly

Technologies: STM32, ESP32, Bluetooth LE, Embedded Linux, Buildroot, QEMU, ArduPilot, MAVLink, GStreamer, Docker, Git, KiCad

Core Skills: Embedded Systems, Bare-Metal Firmware, Device Drivers, FPGA Design, Board Bring-Up, Hardware Validation, RISC-V, PCB Design, Robotics, UAV Systems